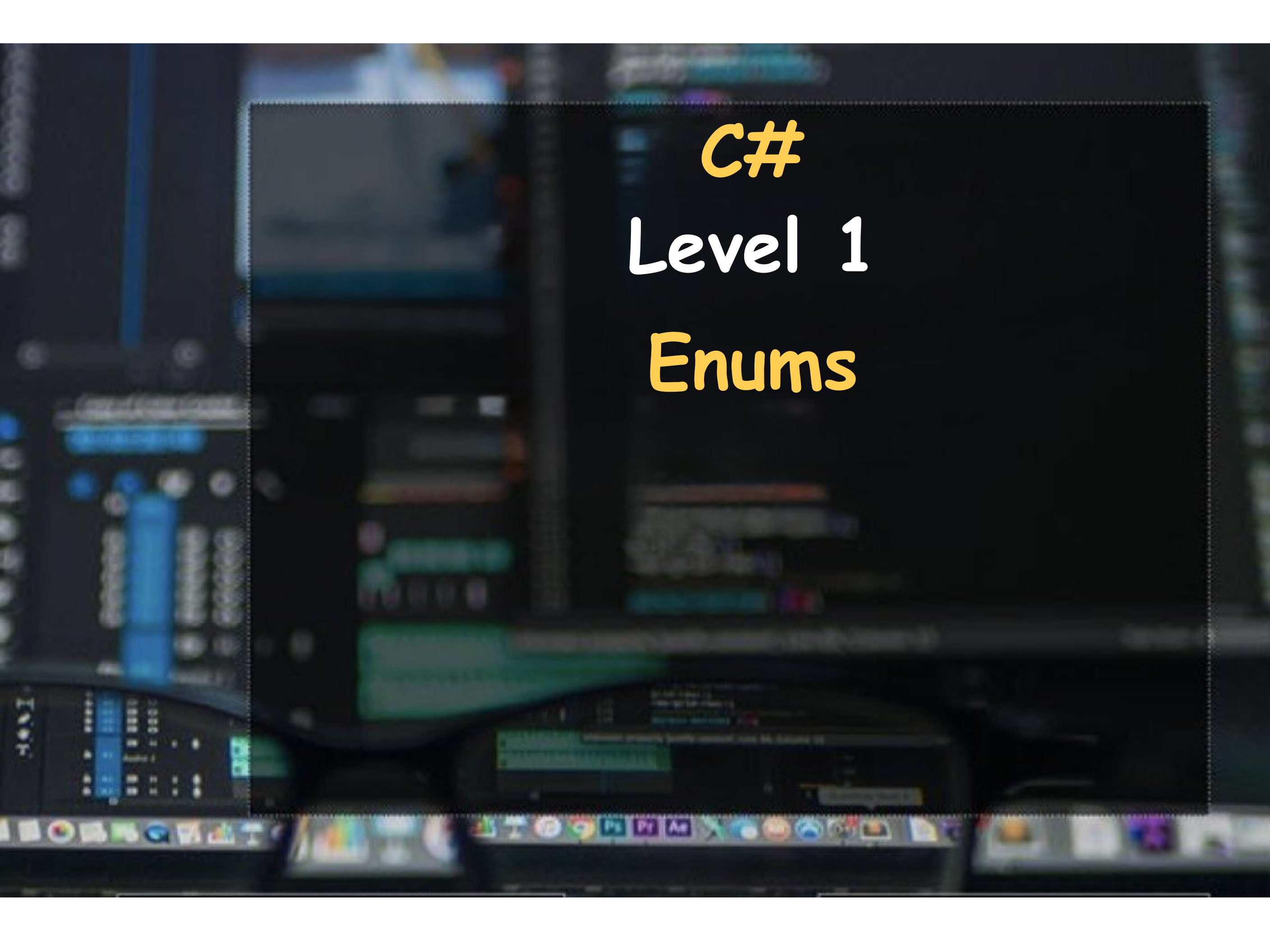
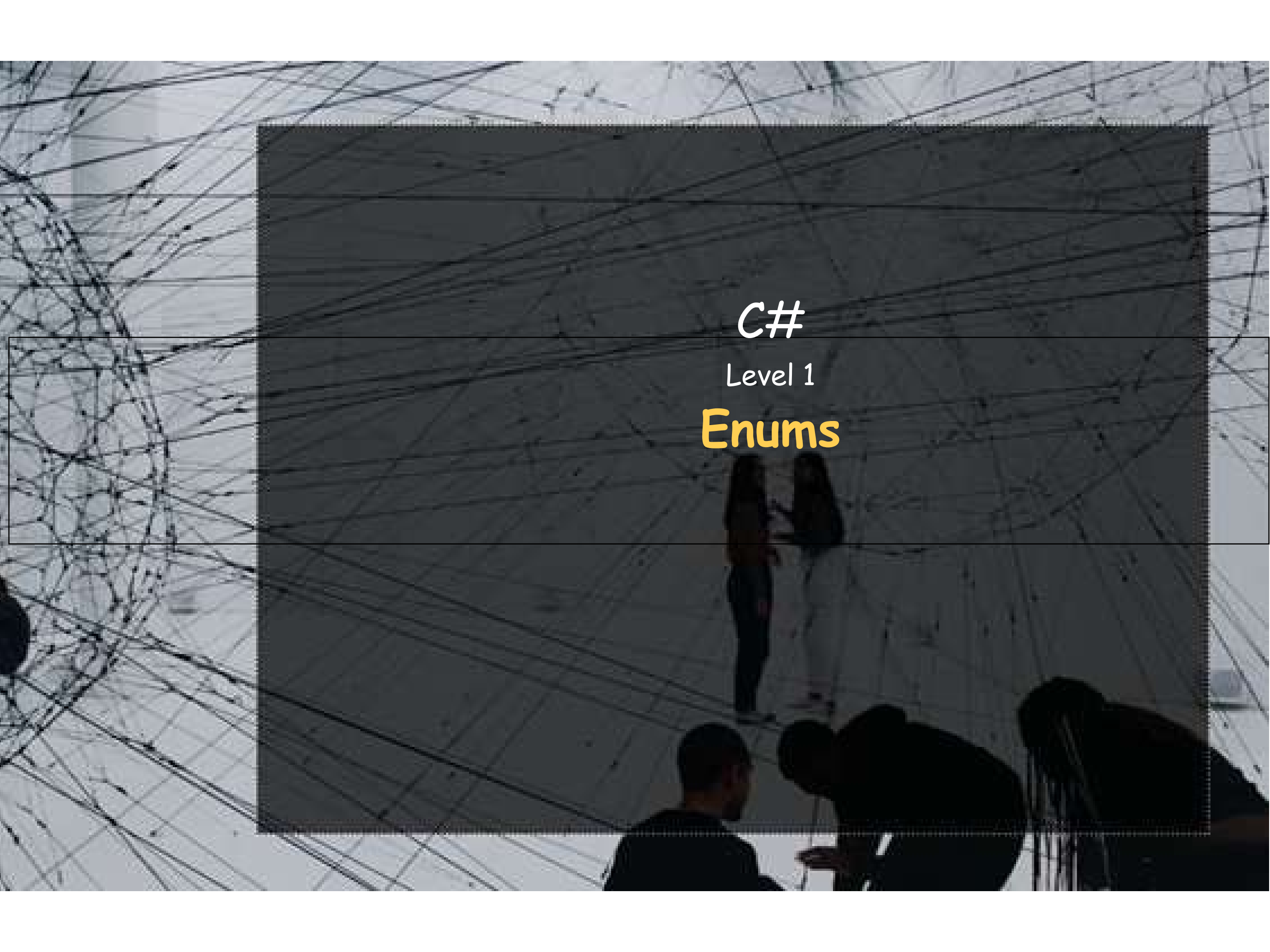


The background is a blurred image of a computer monitor. It shows a dark-themed code editor with various lines of code in different colors (blue, green, white). On the left side, there's a sidebar with some icons and text. At the bottom, a Windows taskbar is visible with several application icons like Chrome, Photoshop, and others.

C# Level 1 Enums



C#

Level 1

Enums

Definition

Enumeration (enum)

-  An `enum` is a type used to define a set of named constant values.
-  Each enum member has an underlying numeric value.

Explanation

Default Numbering

 By default, the first member has the value `0`.

 Each following member increases by `1`.

Procedures

1. Define an enum with named members.
2. Assign numeric values when needed.
3. Declare a variable using the enum type.
4. Assign one of the enum members to the variable.
5. Use or display the selected member.

💡 Example

```
enum enWeekDays
{
    Monday,    // 0
    Tuesday,   // 1
    Wednesday, // 2
    Thursday,  // 3
    Friday,    // 4
    Saturday,  // 5
    Sunday     // 6
}
enWeekDays WeekDay;
WeekDay = enWeekDays.Friday;
Console.WriteLine(WeekDay);
```

➡ Result
Friday

Note

- ◆ Enum members represent named constants, making code easier to read than using unexplained numeric values.
- ◆ If no value is assigned, numbering starts from `0` and increases by `1`.
- ◆ After assigning a value manually, the following members continue from that value.

⊘ Warning

- ◆ The underlying type must be an integral numeric type.

▨ Valid:

```
enum enCategories : byte
```

▤ Invalid:

```
enum enCategories : string
```

- ◆ Be careful when manually assigning values because duplicate numeric values are allowed.



Comparison

▨ Default Enum Values

- ✓ Start at `0`.
- ✓ Automatically increase by `1`.
- ✓ Example: `Monday = 0`, `Tuesday = 1`.

▤ Custom Enum Values

- ✓ Values can be assigned manually.
- ✓ Following members continue from the assigned value.
- ✓ Useful when specific numeric values are required.

✦ Summary

- ✓ ``enum`` defines a set of named constant values.
- ✓ Enum members have numeric values.
- ✓ Default numbering starts at ``0``.
- ✓ Values can be assigned manually.
- ✓ Following members are automatically numbered.
- ✓ The default underlying type is ``int``.
- ✓ Other integral types such as ``byte``, ``short``, and ``long`` can be used.
- ✓ ``string`` cannot be used as an enum's underlying type.

Thank you 😊